Gap Analysis — Offline Bangla Dubbing Video Player

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Summary

Thirteen of the 16 business requirements can be met with existing open-source tools; the work is integration, not invention. Three gaps carry real risk:

1. **Bangladeshi Bangla voice (BR-04):** open Bengali TTS voices are mostly Indian Bengali. Closing this likely needs a Bangladeshi voice dataset and fine-tuning.
2. **Model licences (NFR-08):** some of the best models (NLLB-200, MMS-TTS) are non-commercial. A commercial product needs alternatives.
3. **Live-mode speed (BR-09):** achievable on a good GPU; not on CPU-only machines.

This analysis is based on the BRD dated 2026-10-29. Licence notes reflect general knowledge of these projects and must be re-checked at the source during the spike.

Current vs target state

Aspect	Current state	Target state
How users watch foreign video	Bangla subtitles, or not at all	Spoken Bangladeshi Bangla, no reading needed
Dubbed content available	Only what studios or TV channels dub	Any local video file
Existing AI dubbing	Online, paid per minute, uploads private video	Offline, no fees, nothing leaves the device
Multiple speakers	Single generic voice in most tools	A distinct, consistent voice per speaker
Background audio	Often lost or muffled	Original music and effects kept
Platforms	Mostly web or one OS	Windows, macOS and Linux desktop

Gap per business requirement

Feasibility: **High** = ready tools exist, **Medium** = tools exist but need tuning, **Low** = needs new data or research.

ID	Available today	Gap	Action to close	Feasibility
BR-01 Player	PyQt6 + python-mpv / QtMultimedia	None	Build player shell	High
BR-02 Language detect	faster-whisper auto-detect	Weak on very short or mixed clips	Detect on first 30 s; manual override	High
BR-03 Speech replaced	ASR + translation + TTS chain	No ready end-to-end tool; must integrate	Build pipeline module	High
BR-04 Bangladeshi Bangla	Indian-Bengali voices; translation not dialect-aware	Accent and vocabulary lean Kolkata	Collect Bangladeshi speech data, fine-tune TTS; Bangladeshi glossary or LLM post-edit	Low
BR-05 Voice per speaker	pyannote.audio diarization; few Bengali voices	Limited voice variety; diarization errors on overlap	Voice presets + pitch/speed variation; merge-split tools	Medium
BR-06 Keep background	Demucs vocal separation	Slow on CPU; some artefacts	GPU when available; lighter model option	High
BR-07 Fully offline	All models run locally	First-run download needed	Model manager with resume and checksums	High
BR-08 Pre-process mode	All components	Bangla lines often longer than original	Time-stretch, shorten translation when too long	High

ID	Available today	Gap	Action to close	Feasibility
BR-09 Live mode	Same components	Pipeline latency of several seconds per line	Rolling buffer, parallel workers, GPU only	Medium
BR-10 Choose voices	TTS presets	No UI	Speaker-voice mapping panel	High
BR-11 Switch audio	mpv multiple audio tracks	None	Add track toggle	High
BR-12 Export	ffmpeg muxing	None	Export dialog	High
BR-13 Subtitles	Translation output	None	Write SRT, show in player	High
BR-14 Cache	Local storage	None	Cache keyed by file hash	High
BR-15 Edit lines	TTS can regenerate single lines	No editor UI	Line editor panel	High
BR-16 Bangla UI	Qt translation system	Translation work	Qt Linguist strings	High

Technology and licence gaps

The licence column is from general knowledge and must be verified at each project's source before any release decision.

Component	Candidate	Licence (to verify)	Commercial use	Gap / note
Speech-to-text	faster-whisper (Whisper large-v3 / turbo)	MIT	Yes	None
Diarization	pyannote.audio 3.x	MIT (gated model download)	Yes	Needs a Hugging Face token once
Separation	Demucs	MIT	Yes	Slow on CPU

Component	Candidate	Licence (to verify)	Commercial use	Gap / note
Translation	NLLB-200	CC-BY-NC 4.0	No	Non-commercial only
Translation	IndicTrans2	MIT	Yes	Good Bengali; check source-language coverage
Translation	Local LLM (e.g. Qwen) via llama.cpp	Varies by model	Varies	Most natural Bangla; heavier on hardware
Bangla TTS	Indic Parler-TTS	Apache 2.0	Yes	Indian-Bengali accent; several voices
Bangla TTS	MMS-TTS (ben)	CC-BY-NC 4.0	No	Single voice
Voice cloning	XTTS-v2	Coqui Public Model Licence	No	No Bengali support
Player	PyQt6 / python-mpv	GPL or commercial (Qt); mpv LGPL/GPL	Conditional	PySide6 (LGPL) is an option if a closed-source app is needed

For a commercial product, the licence-safe stack is: faster-whisper, pyannote, Demucs, IndicTrans2, Indic Parler-TTS, PySide6 or PyQt6 with a commercial licence.

Requirement analysis

Refined requirements

ID	Original wording	Refined, testable wording
BR-03	Replace speech with spoken Bangla	Every detected speech segment of 1 s or more has a Bangla audio line; untranslated segments fall back to original audio

ID	Original wording	Refined, testable wording
BR-04	Bangladeshi vocabulary and pronunciation	70%+ of Bangladeshi UAT testers rate the voice as "sounds Bangladeshi" (target to confirm)
BR-05	Different consistent voice per speaker	Up to 6 speakers per video; the same speaker keeps the same voice across the whole file
BR-09	Live mode within 60 s	On the recommended GPU, playback starts within 60 s and never pauses for buffering on a 2-hour film

Dependencies (build order)

1. Audio extraction (ffmpeg) is needed by everything else.
2. Speech-to-text feeds translation; diarization feeds voice assignment.
3. Translation feeds TTS; TTS feeds timing fit; timing fit and separation feed the mix.
4. Pre-process mode (BR-08) must work before live mode (BR-09).
5. Voice selection, line editing and export build on the finished pre-process pipeline.

Working assumptions for the BRD's open questions

These stand until the product owner answers.

Open question	Working assumption
Commercial, free or internal?	Treat as commercial; choose only licence-safe models
First source languages	English, Hindi, Korean, Arabic
Minimum hardware	8 GB NVIDIA GPU or Apple Silicon 16 GB for live; CPU only for pre-process
Bangladeshi voice data	Not available; plan to record or license 10+ hours
Buffer and preset voices acceptable?	Yes, 30–60 s buffer and preset voices
Budget and release date	Unknown; needed before planning development
Team names	Unknown; needed before planning development

Decisions needed and spike goals

Decisions for the product owner

- [ ] Commercial or non-commercial release (decides the model stack)
- [ ] Accept Indian-Bengali voices for version 1, or fund Bangladeshi voice data first
- [ ] Confirm minimum supported hardware
- [ ] Confirm the proposed success targets in the BRD

What step 3 (spike / feasibility) must prove

#	Question	Pass condition
1	Does the end-to-end chain work offline?	A 5-minute English clip dubs into Bangla with no network
2	Which translation sounds most natural?	IndicTrans2 vs NLLB vs local LLM, scored by 3+ Bangladeshi reviewers
3	How Bangladeshi do the TTS voices sound?	Reviewer score recorded for each candidate voice
4	Does diarization hold up?	90%+ correct speaker labels on a 3-speaker clip
5	How fast is it?	Processing time per minute of video measured on GPU, Apple Silicon and CPU
6	Does timing fit work?	Dubbed lines start within 0.5 s of the original

If the spike fails on 1, 4 or 6, return to step 1 for rework. If it fails only on 3, continue with preset voices and plan voice fine-tuning as a later phase.